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NPS/002 /014 – Specification for LV Joints and Terminations

1. Purpose

This Specification details requirements for low voltage cable joints and terminations in "kit form" for use on the company's distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/014	1	Specification for LV Joints and Terminations

2. Scope

This document applies to the specification required and describes the requirements of CE Electric UK, with respect to: -

- Mains and Service Straight Joints
- Mains and Service Branch Joints
- Mains and Service Stop Ends
- Pole and LV board Termination

Joints, stop ends and terminations may be either cold applied or require the application of heat.

3. Technical Requirements

3.1 Conditions of Installation.

Joints and terminations specified in this document shall be installed on cables laid in accordance with the CE Electric UK policy for the installation of cables: DSS/026. During storage and after installation cable joints can be and are subjected to the full range of climatic conditions encountered in the UK.

Cable joints may be surrounded by ground water for most of its operating life. Where cable is installed in ducts, flooding of ducts can occur resulting in permanently wet sections along the cable route.

Cable joints installed above ground will be supported by means of cleats either vertically or horizontally and these cables may be exposed to direct sunlight for significant periods. Cable terminations will be installed up in and in contact with the wood poles and therefore in contact with a pole preservation medium such as creosote.

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3.2 Conditions of Operation for Cable Joints

LV joints and terminations purchased in accordance with this specification are required to operate under conditions stipulated in IMP/001/911. The following are the general conditions under which cable joints purchased in accordance with this Specification are required to operate:

- Nominal system voltages: 400/230 volts.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 10%.
- Nominal system frequency: 50Hz
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points.

3.3 Components

Components specified shall be suitable for use with impregnated paper insulated cables including those to BS 6480, and polymeric insulated cables including those to BS 6004, BS 6007, BS 5467, BS 6346, HD 603, ENATS 09-7 and 09-9. They shall not be adversely affected by contact with other materials normally used in the construction of cable joints, terminations or stop ends, and shall not increase the rate of corrosion of any metal with which they come into contact.

Assembled components forming part of a cable system shall be capable of operating under the normal and fault temperature conditions specified in the relevant cable specifications.

Mechanical connections to the metallic earth screen of the cable shall be tested in accordance with engineering recommendation C93.

All connectors and lugs shall be in accordance with Engineering Recommendation C79 or BS EN 61238:1.

3.4 Low voltage Joints

Insulation Piercing along with mechanical phase and neutral/earth connectors shall be used within the relevant joint kits.

The rated voltage of the components covered by this specification is as defined in BS7888. All components must have at least an equal current rating to the cables to which they are connected.

Current ratings shall be calculated in accordance with IEC 287, using standard UK ground conditions and an installation depth of 450mm.

3.5 Joint shell and Resin

For resin filled joints, the joint shell shall be horizontally split and a minimum 1.5mm thick plastic or an alternate material, which shall be approved by CE Electric UK. The shell shall have stepped cable entry ports for varying cable diameters (as detailed in appendix 1). The joint shell shall be compatible with a cold pour natural or synthetic oil based polyurethane resin and so designed to allow at a minimum 10mm encapsulation at every point within its internal length.

The shell assembly shall be supplied complete with sealing clips/strips and a means of sealing the cable entry ports.

3.6 Cable Range & Joint Requirements

The most common types of cables that will be encountered on the distribution system and a range of joints required are as detailed in Appendix 1.

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3.7 Low Voltage Terminations

In general terminations will be carried out on 3-phase, CNE, XLPE insulated concentric copper wire waveform neutral/earth with solid aluminium phase conductors and PVC oversheath to BS 7870.

Terminations are required for the following equipment.

- ☐ Low Voltage Fuse Boards (conductor size 185mm² and 300mm²)
- ☐ Industrial Service Unit (conductor size 95mm² 185mm² and 300mm²)
- ☐ Low Voltage Overhead lines (conductor size 95mm², 185mm² and 300mm²)

3.8 Stop ends

Stop ends, which shall be supplied in kit form, are required for each cable type and size as detailed in Appendix 1. The stop end kits shall contain all the necessary components to carry out the joint.

Mechanical connections to the metallic earth screen of the cable shall be tested in accordance with engineering recommendation C93.

All connectors and lugs shall be in accordance with Engineering Recommendation C79 or BS EN 61238:1.

When resin free type stop ends are proposed they will meet the requirements of the relevant parts of Engineering Recommendation C81/3, any heat shrinkable materials used shall be in accordance with ENATS 09-11.

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4 References

4.1 External Documentation

The Cable Joints shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

4.2 British Standards

Reference	Title
BS 7888	LV and MV accessories for power cables with rated voltage from 0.6/1kV (Um = 1.2 kV) up to and including 20.8/36 kV (Um = 42kV)
BS EN 50266	Tests on electric cables under fire conditions
BS 6004	PVC insulated electric cables, non armoured, up to and incl 450/750v.
BS 6007	Electric cables unsheathed, heat resisting, up to and incl 450/750v.
BS EN 61238-1	Compression and mechanical connectors for power cables.
BS 5467	Specification for 600/1000 V and 1900/3300 V armoured electric cables having thermosetting insulation
BS 6234	Specification for polyethylene insulation and sheath of electric cables
BS 6346	Specification for 600/1000 V and 1900/3300 V armoured electric cables having PVC insulation
BS 6360	Conductors in Insulated Cables and Cords
BS 6480	Specification for impregnated paper-insulated lead or lead alloy sheathed electric cables of rated voltage up to and including 33000 V
BS 6724	Specification for 600/1000 V and 1900/3300 V armoured electric cables having thermosetting insulation and low emission of smoke and corrosive gases when affected by fire
BS 7870 3.40	LV and MV polymeric insulated cables for use by distribution and generation utilities. Specification for distribution cables of rated voltage 0.6/1 kV. XLPE insulated, copper wire waveform concentric cables with solid aluminium conductors
BS 7889	Specification for 600/1000 V single core un-armoured electric cables having thermosetting insulation

4.3 Energy Networks Association Technical Specification

Reference	Title
ENATS C93	Type approval tests for mechanical connections to metallic sheaths of cables.
ENATS 09-7	PVC & XLPE Insulated and Concentric Service cables.
ENATS 09-9	XLPE Insulated CNE Cable with Solid Al Phase Conductors & Concentric Waveform Al Neutral Earth.
ENATS 09-11	Heat Shrinkable Material for use on 600/1000v Cables & Accessories.
ENATS-C79	Type Approvals tests for Connectors and Terminations for Aluminium conductors of insulated power cables.
ER-C81/3	Type Approvals tests for Joints for 600/1000v Cable Systems.

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4.4 Other

Reference	Title
HD 603	Distribution Cables of rated Voltage 0.6/1 kV.
IEC 287	Calculation of Current Ratings.

4.5 Internal documentation

Reference	Title
IMP/001/911	CoP for Economic Development of LV Networks
DSS/026	Policy for the Installation of Distribution Power Cables

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	23/09/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	29/09/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	08/10/08

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	22/10/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	17/10/08

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	20/10/08

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Appendix 1 – Cable Range and Joint Requirements

Mains Cables Type

1. 3-Core CNE / 4-Core SNE XLPE insulated concentric with aluminium or copper wire waveform neutral/earth conductors, PVC oversheath cables with solid aluminium phase conductors to BS7870.
CNE Conductor sizes: - 70mm², 95mm², 120mm², 185mm² and 300mm²
SNE Conductor sizes:- 95mm², 185mm² and 300mm²
2. 4-Core Paper insulated, lead sheathed steel tape armoured served cables with stranded aluminium or stranded copper conductors to BS 6480. Conductor sizes up to and including 0.3in².
3. Impregnated paper-insulated cables with aluminium sheath/neutral conductor and three shaped solid aluminium phase conductors (CONSAC)
Conductor sizes: - 70mm², 120mm², 185mm² and 240mm²

Service cables Type

4. 4-Core Paper insulated, lead sheathed steel tape armoured served cables with stranded aluminium or stranded copper conductors to BS 6480.
Conductor sizes up to and including 0.06in²
5. Concentric CNE / SNE Single and three-phase PVC insulated cable with solid aluminium phase conductors and stranded copper neutral.
Conductor size 25mm² and 35mm²
6. Concentric SNE PVC insulated cable with stranded Copper conductors.
Conductor range 4mm², 16mm² and 25mm²

The table below detail's range of joints required.

Joint	First Cable	Second Cable	Joint Type
Mains to Mains	Type 1	Type 1	Resin Filled
Mains to Mains	Type 1	Type 2	Resin Filled
Mains to Mains	Type 2	Type 2	Resin Filled
Mains to Mains	Type 1	Type 3	Resin Filled
Mains to Service	Type 1	Type 5	Resin Filled / Resin Free
Mains to Service	Type 2	Type 5	Resin Filled / Resin Free
Mains to Service	Type 3	Type 5	Resin Filled / Resin Free
Mains to Double Service	Type 1	Type 5	Resin Filled / Resin Free
Mains to Double Service	Type 2	Type 5	Resin Filled / Resin Free
Service to Service	Type 4	Type 4	Resin Filled / Resin Free
Service to Service	Type 4	Type 5	Resin Filled / Resin Free
Service to Service	Type 4	Type 6	Resin Filled / Resin Free
Service to Service	Type 5	Type 5	Resin Filled / Resin Free
Service to Service	Type 5	Type 6	Resin Filled / Resin Free
Service to Service	Type 6	Type 6	Resin Filled / Resin Free

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Appendix 2 - Addendum to Supplier requirements

The Joint Kit shall incorporate all the necessary components (excluding the resin) to carry out a specific joint combination. As and when required additional "module kits" shall be supplied.

When resin free type joints are proposed, they will meet the requirements of the relevant parts of Engineering Recommendation C81/3. Any heat shrinkable materials used shall be in accordance with ENATS 09-11.

Each individual joint, termination and stop end kit shall include the relevant jointing instruction/drawing and a kit content list. Resin volume (where applicable) shall clearly be displayed.

The production of the instruction etc. is the responsibility of the supplier, and any alterations shall be subject to discussions between the supplier and CE Electric UK.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

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Appendix 3 - logistical requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK's product code
- * Weight

Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.

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4 Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

LV joints and terminations required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

4 Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE: Applicable sections shall be completed for each size / range of joints and connectors offered.

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		Clause / Requirements	Conformance Code	Remarks / Comments
Joint Shells: - Horizontally Split - Min 1.5mm thick Plastic or equivalent Material - Stepped Cable Entry Ports - Range of Sizes (to fit cables as specified in Appendix 1) - Compatible with Cold Pour or Synthetic Oil Based Polyurethane Resin - Provide min 10mm Encapsulation - To be supplied with approved JI's - Supplied with suitable Seal Clips / Strips		NPS/002/004 Parts 3.5 & 3.8		

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	Clause / Requirements	Conformance Code	Remarks / Comments	
Mechanical Connections to Metallic Sheaths of Cables:				
Test Specimens	BS EN 61238-1 Part 5			
Initial Resistance Checks	C79 Part 5			
	C93			
Short Circuit Tests	C79 Part 6			
	BS EN 61238-1 6.3.4			
	C93			
Load Cycling Tests	C79 Part 7			
	BS EN 61238-1 Part 6.5			
	C93			
Tensile Tests	C79 Part 8			
	BS EN 61238-1 Part 7			
Heat Shrink Materials:				
Visual Examination	ENATs 09-11 Test A			
Dimensions & Longitudinal Shrinkage	ENATs 09-11 Test B			
Tensile Strength & Ultimate Elongation	ENATs 09-11 Test C			
Heat Shock-Tubing	ENATs 09-11 Test D			
Heat Shock-Moulded Parts	ENATs 09-11 Test E			
Hot Modulus	ENATs 09-11 Test F			
Electrical Strength	ENATs 09-11 Test G			
Secant Modulus	ENATs 09-11 Test H			
Flame Retardant	ENATs 09-11 Test J			
Water Absorbance	ENATs 09-11 Test K			
Low Temperature Flexibility	ENATs 09-11 Test L			
Stiffness	ENATs 09-11 Test M			
Heat Ageing	ENATs 09-11 Test N			

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	Clause / Requirements	Conformance Code	Remarks / Comments
Corrosion Resistance	ENATs 09-11 Test O		
Solvent Resistance	ENATs 09-11 Test P		
Adhesive Peel Strength	ENATs 09-11 Test Q		
Low Temperature Flexibility of Adhesive	ENATs 09-11 Test R		
Flow of Adhesive at Elevated Temperature	ENATs 09-11 Test S		
Weather Resistance	ENATs 09-11 Test T		